

GESTURE MASTER

CprE 186

Section I Group 3

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PROBLEM

- Create a device that interprets hand motions
- Acquire an understanding of how hardware and software interact to innovate and solve real-world problems.
- Designing a system that could serve as a base for larger-scale motion interpretation (ASL, etc.)

SOLUTION

- Utilized an Adafruit BNO055 gyroscope and accelerometer sensor, Arduino Uno, and the Arduino IDE
- Learned hardware functionality
- Learned how sensor hardware data is interpreted within the IDE
- Performed numerous data analyses for accuracy

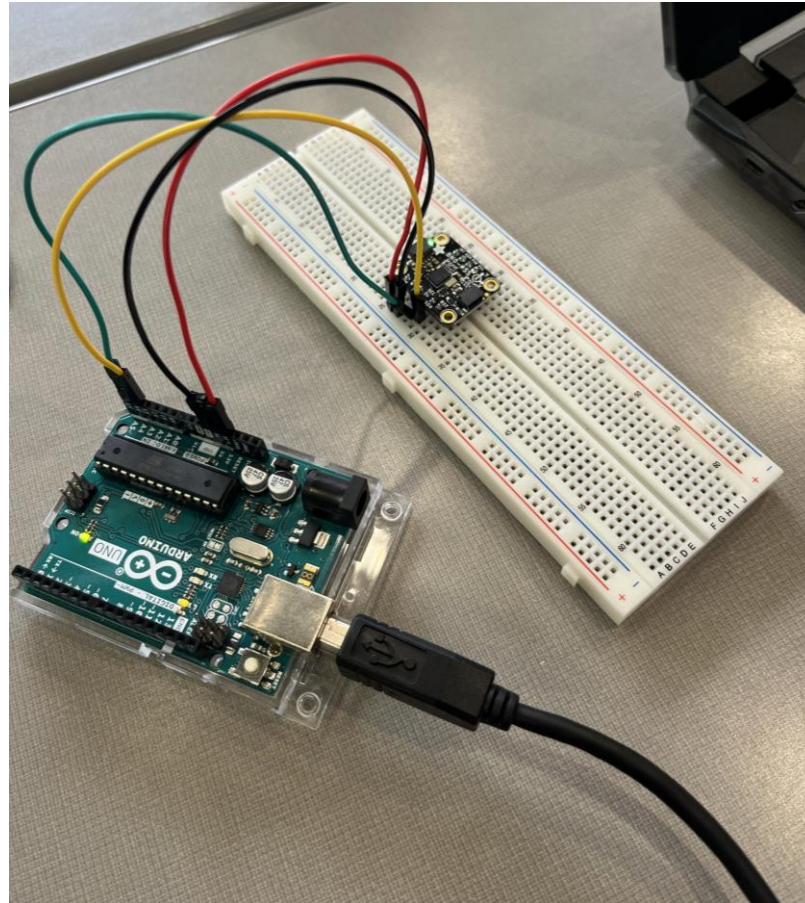
USERS & USES

- Anyone!
- Disability assistance via gesture control
- Gesture-based digital display interaction
- Aiding mobility-limited healthcare tech interaction
- Entertainment and interactive learning

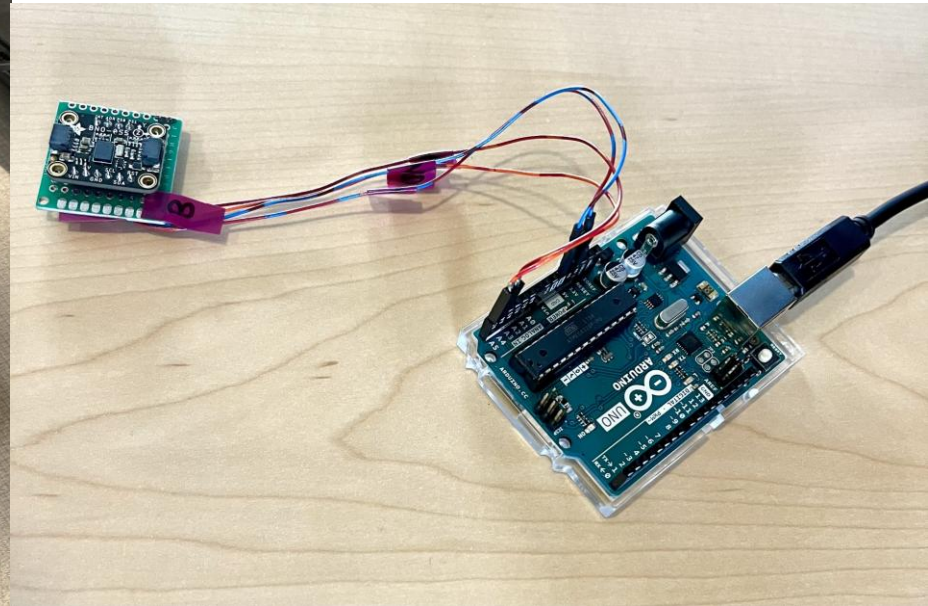
DESIGN REQUIREMENTS

- Retrieve gyroscope and accelerometer data for analysis
- Evaluate vector data
- Determine type of motion being performed
- Keep track of the motion time
- Display appropriate (humorous) output

PROJECT PHOTOS



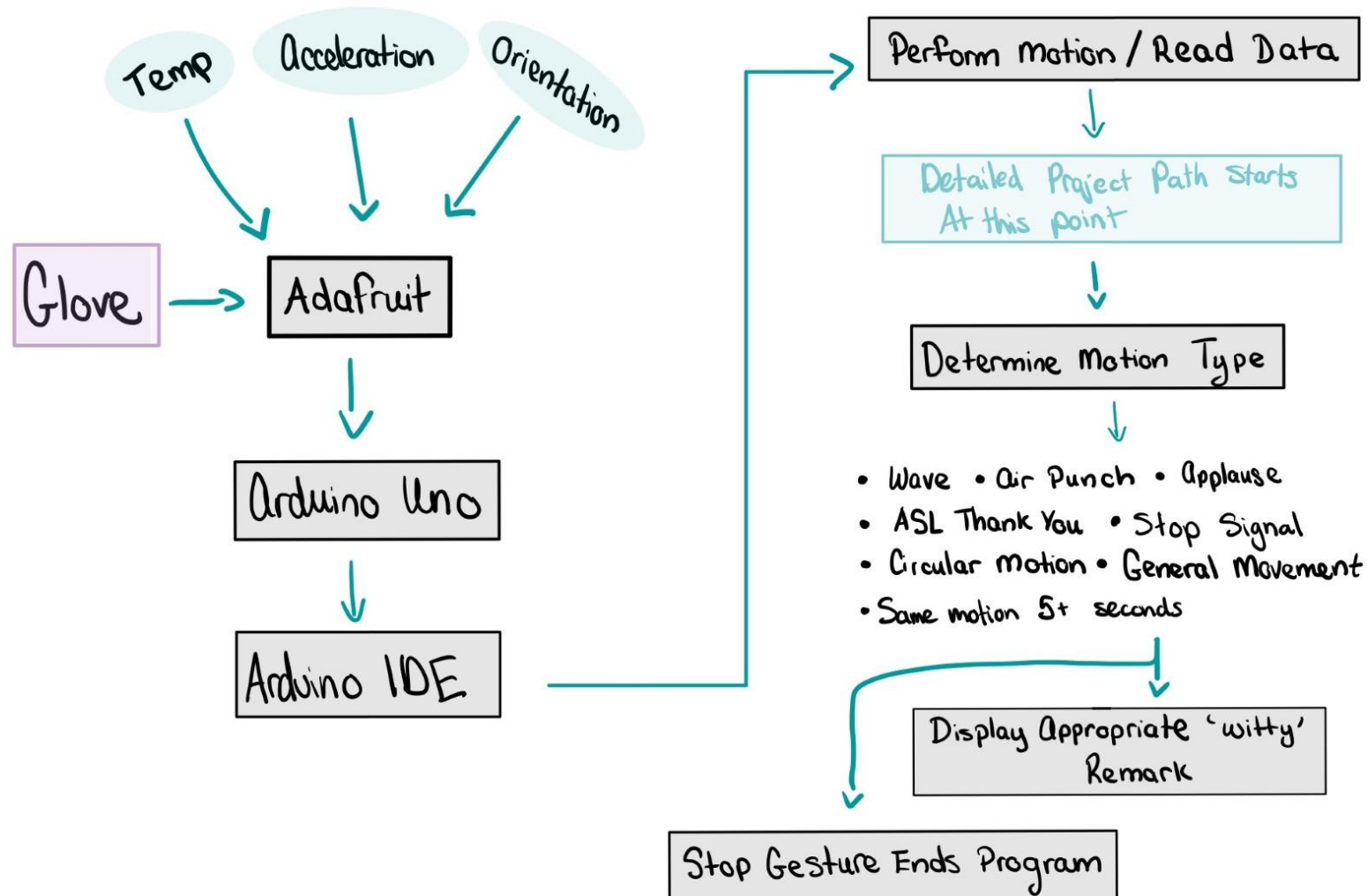
First draft of the hardware for
our project



Final soldered version of
project hardware

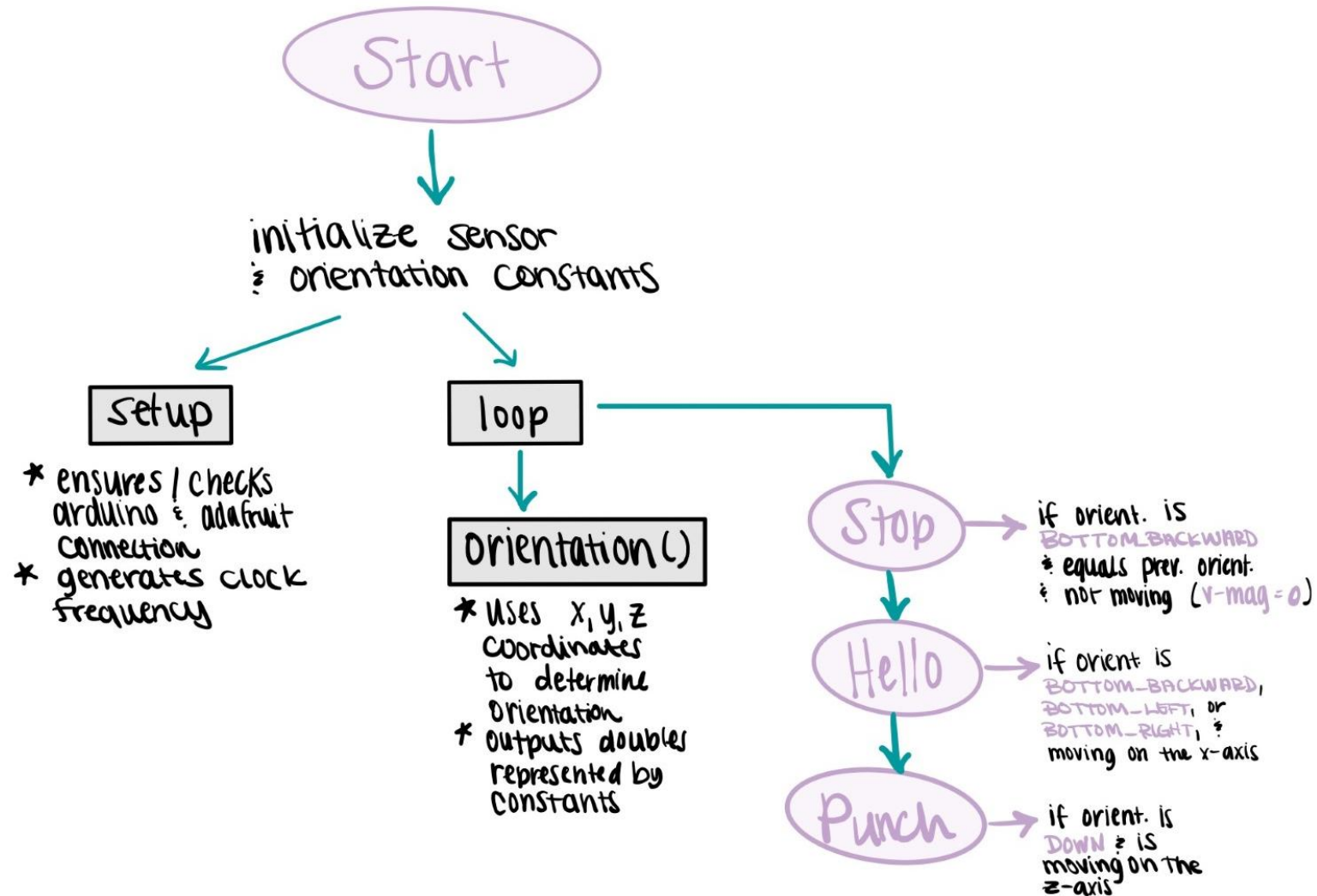
DESIGN APPROACH (UML DIAGRAM)

General Project Path



SOFTWARE (UML DIAGRAM)

Detailed Software Process



PROJECT RESOURCES

Wiring for Arduino:

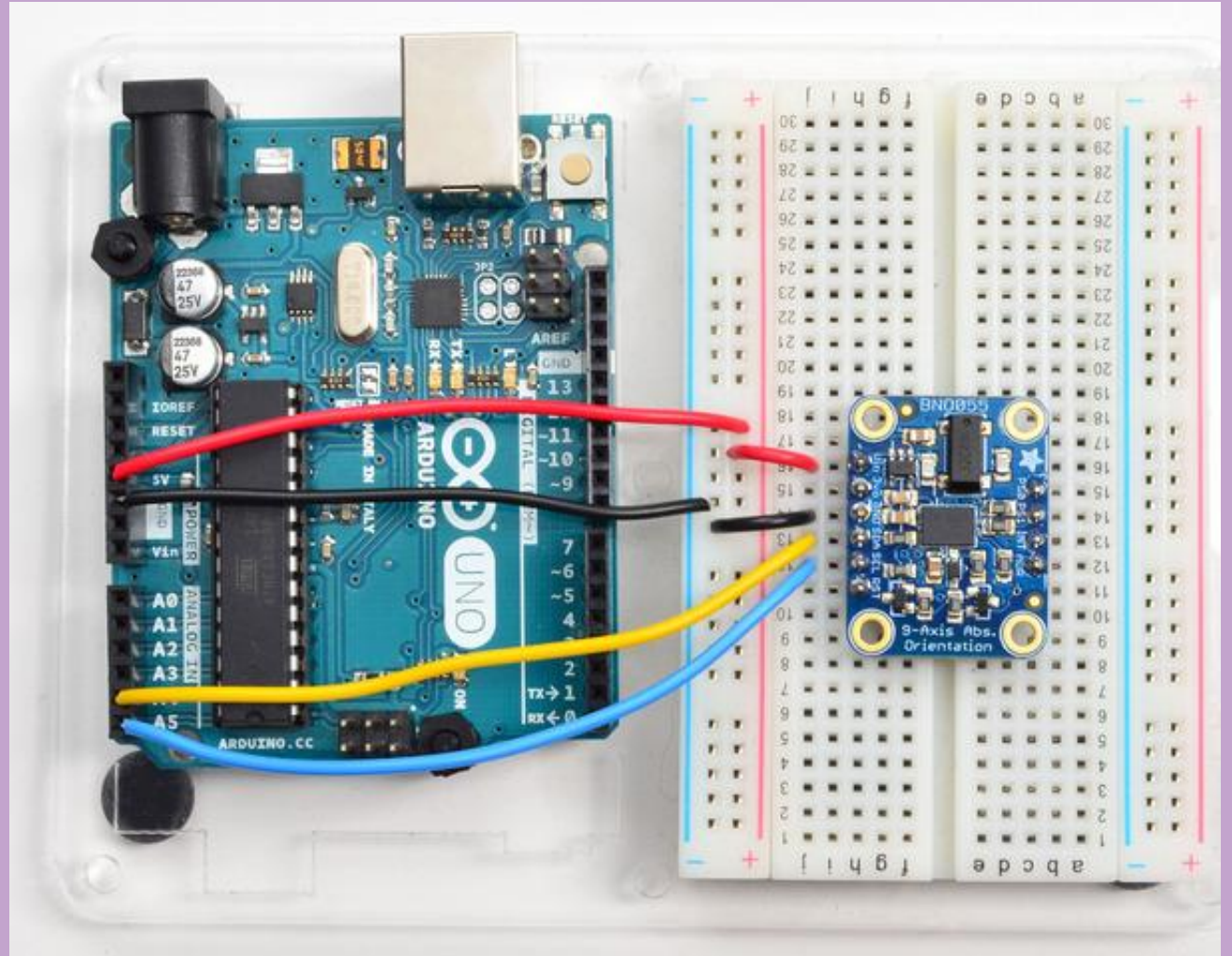
<https://learn.adafruit.com/adafruit-bno055-absolute-orientation-sensor/arduino-code>

Adafruit Set Up:

<https://learn.adafruit.com/adafruit-arduino-ide-setup/overview>

Adafruit Sensor in Detail:

<https://learn.adafruit.com/using-the-adafruit-unified-sensor-driver/how-does-it-work>



**THANK
YOU!**

